

Descriptive Statistical Analysis of Audience Reviews: A Case Study of 3 Idiots

Kingshuk Goswami

June 3, 2026

Overview

- 1 Project Overview
- 2 Dataset Creation
- 3 Research Questions
- 4 Descriptive Statistics
 - Central Tendency
 - Range Analysis
 - Standard Deviation & Variance Analysis
 - Skewness Analysis
- 5 Interval Frequency Distribution
- 6 Statistical Summary
- 7 Research Questions Answered
- 8 Audience Effort Dynamics

Project Overview & Movie Introduction

Project Objective

The objective of this project is to analyze audience reviews of the movie 3 Idiots collected from IMDb using descriptive statistical techniques. By combining statistical calculations with graphical visualizations, the project aims to provide meaningful insights into the movie's popularity and overall audience reception.

Film Profile

Title: 3 Idiots

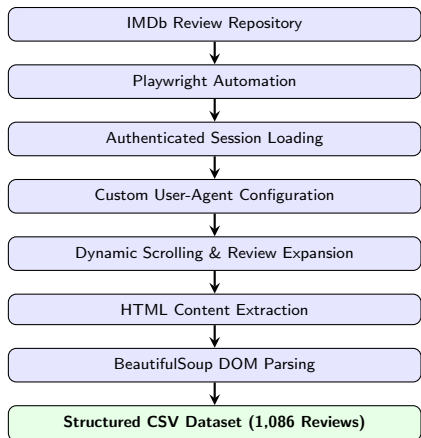
Release Year: 2009

Director: Rajkumar Hirani

Genre: Comedy-Drama

Runtime: 2hr 50mins

Dataset Creation Pipeline



Pipeline Mechanics

- **Dynamic Content Loading:** Automated scrolling retrieves dynamically rendered reviews
- **Session-Based Access:** Stored login sessions maintain consistent page access.
- **Efficient Parsing:** BeautifulSoup extracts review metadata and text fields.

Dataset Validation Summary

- Raw Dataset Size: **1086 Reviews**
- Clean Dataset Size: **1042 Reviews**
- Duplicate Reviews Removed
- Non-Numeric Ratings Removed
- No Missing Ratings
- Retention Rate: **96%**

Outcome

A structured, labelled dataset of 1042 validated reviews was retained for descriptive statistical analysis.

Research Questions

Key Questions Investigated

- 1 Is **3 Idiots** generally liked by audiences?
- 2 Are audience opinions consistent or highly divided?
- 3 Which rating values and intervals dominate audience opinion?
- 4 What does the rating distribution reveal about viewer behavior?
- 5 Do review-writing patterns support the statistical findings?

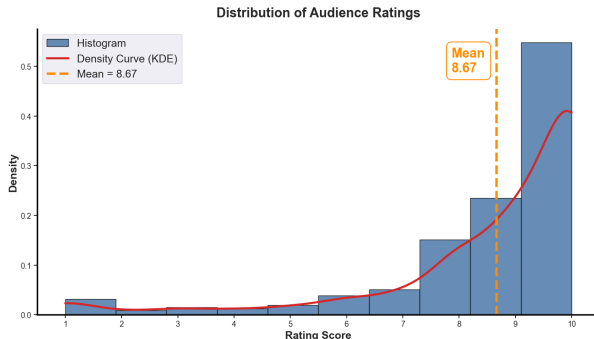
Project Goal

Use descriptive statistics and audience reviews to scientifically evaluate the popularity and reception of **3 Idiots**.

Arithmetic Mean Analysis

Strong Approval of Audience

The average audience rating of 8.67/10 indicates exceptionally strong audience satisfaction. The average rating is substantially above the scale midpoint, indicating a strongly positive audience response. Most viewers perceived the film as both engaging and impactful, reflecting broad appreciation across different audience groups.

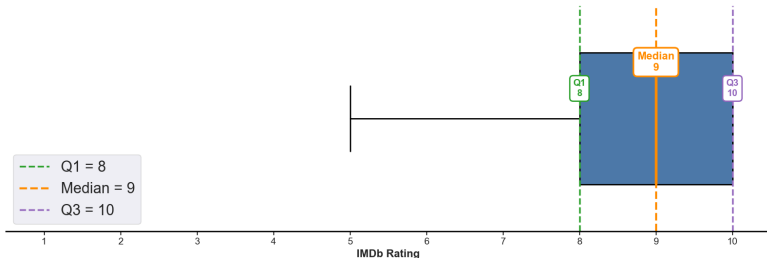


Median Analysis

Most Viewers Rated Highly

Half of all reviewers rated the movie 9 or higher. The median rating confirms that favorable reviews are widespread across the audience rather than concentrated among a small group of highly satisfied viewers. Unlike the mean, the median is less affected by the outliers, making it a reliable representation of the typical viewer's opinion.

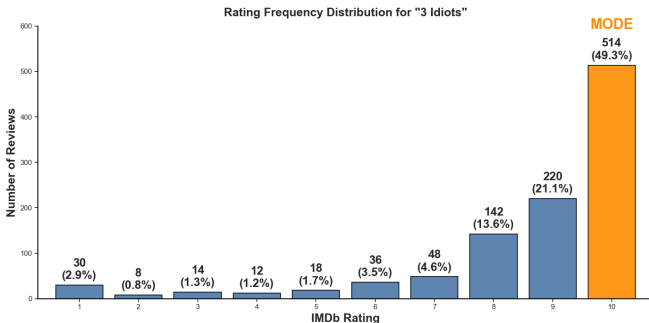
Median Analysis: Central Audience Opinion



Mode Analysis

Perfect Scores is the Most Common Opinion

Nearly half of all reviewers awarded a perfect 10/10, making excellence the dominant audience opinion. The most frequently occurring rating is the maximum possible score, showing exceptional audience satisfaction and a highly memorable viewing experience. A perfect score was the dominant audience opinion.



Range Analysis

Diverse Opinions

The movie received ratings across the entire scale, from 1 to 10. The mean, median, and mode all indicate that 3 Idiots was highly appreciated by most viewers, reflecting strong overall audience satisfaction. However, the full rating range reveals that not everyone shared this opinion, with a small minority giving very low scores, showing that some viewers criticized or even strongly disliked the film despite its widespread acclaim.

Range Analysis: Audience Ratings Across the IMDb Scale



Range = 9

Min = 1

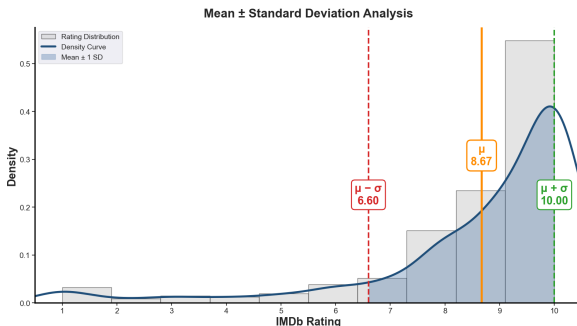
Max = 10



Standard Deviation & Variance Analysis

Audience Consensus Remains Strong

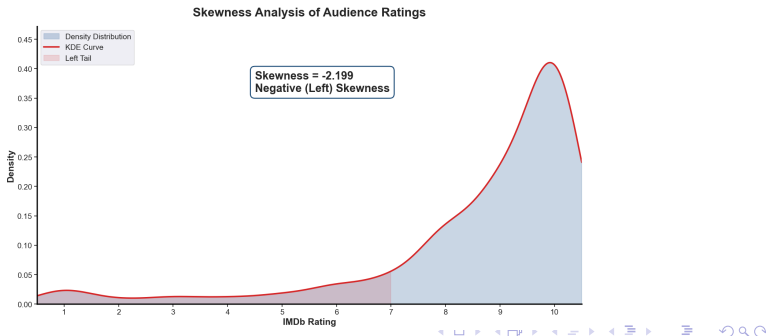
Despite individual differences in ratings, audience opinion remains largely consistent. The standard deviation of 2.07 indicates moderate variation around the average rating of 8.67, suggesting that while viewers did not rate the movie identically, most ratings remained concentrated in the positive range.



Skewness Analysis: Distribution Asymmetry

Ratings Concentrate Near Excellence

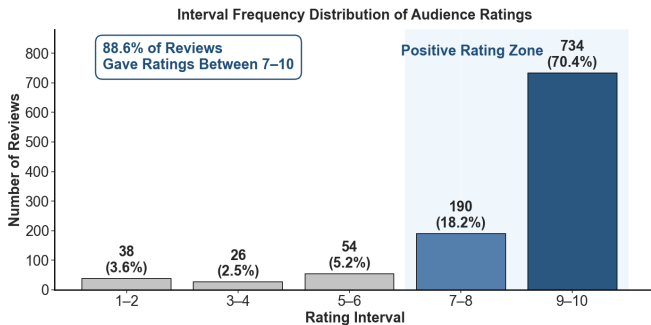
Ratings are heavily concentrated at the high end of the scale. The audience overwhelmingly leaned toward very high ratings. The rating distribution is concentrated near the upper end of the scale that indicates strong audience approval and is commonly observed in films that receive widespread acclaim.



Interval Frequency Distribution

High-Rating Intervals Dominate

The vast majority of ratings fall in the positive zone. The frequency distribution shows that ratings are heavily concentrated in the 7-10 range, with nearly 9 out of every 10 reviews falling in the 7-10 range. Neutral and negative ratings constitute only a small minority, indicating that audience appreciation was not merely positive but overwhelmingly enthusiastic.



Statistical Dashboard

Metric	Value	Interpretation
Mean	8.67	Strong audience approval.
Median	9	More than 50% viewers rated highly.
Mode	10	Perfect score dominates.
Range	9	Diverse opinions exist.
Standard Deviation	2.07	Strong audience consensus.
Skewness	-2.199	Ratings near excellence.
Ratings Between 7–10	88.6%	9 of 10 positive reviews.
Perfect 10/10 Ratings	49.3%	Nearly half gave 10/10.

One-Line Verdict

Strong approval, broad consensus, and ratings concentrated near excellence.

What Do All Rating Metrics Tell Us Together?

Insight 1: Popularity Was Broad-Based Rather Than Isolated

The film's strong reputation is not the result of a small group of enthusiastic viewers. The close alignment of the mean (8.67), median (9), and mode (10), together with 88.6% of ratings falling between 7 and 10, shows that positive sentiment was shared across the audience. The strong negative skewness further confirms that ratings were concentrated near the highest end of the scale.

Insight 2: Criticism Existed but Did Not Alter the Audience Verdict

Ratings span the entire scale from 1 to 10, indicating that some viewers strongly disagreed with the majority opinion. However, the high average rating (8.67) and moderate standard deviation (2.07) show that these negative opinions remained relatively uncommon. As a result, the overall audience consensus remained strongly favorable despite the presence of criticism.

Research Questions Answered

Research Question	Finding
Is the movie generally liked?	Mean = 8.67 and Median = 9 indicate strong audience approval.
Is it controversial?	Ratings span 1–10, but negative ratings remain uncommon.
What does skewness reveal?	Ratings are concentrated near the highest scores.
Which interval dominates?	The 9–10 interval contains 70.4% of all ratings.
Do graphs support statistics?	All visualizations consistently support the numerical findings.

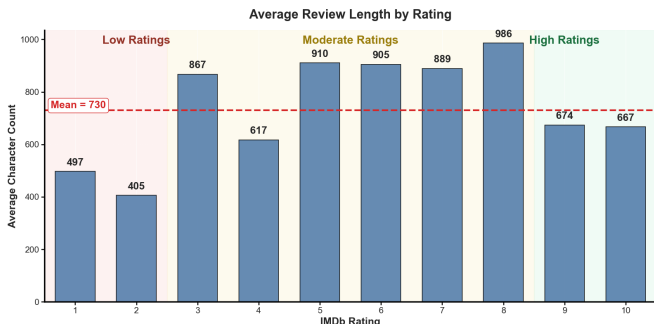
Overall Verdict

3 Idiots received strong, widespread, and statistically consistent audience approval.

Audience Effort Dynamics

Mixed Opinions Need Justification

Viewers with mixed opinions write nearly twice as much as viewers with extreme opinions. Viewers assigning moderate ratings consistently wrote longer reviews, indicating a greater need to explain mixed opinions. In contrast, reviewers extreme ratings tend to write shorter reviews, suggesting that strong positive or negative judgments are expressed with greater certainty and require less justification.



Final Conclusions

The Statistical Verdict

The analysis confirms that **"3 Idiots"** received **strong, widespread, and consistent audience approval**. High central tendency measures, concentration of ratings in the 7–10 range, and strong negative skewness collectively indicate that positive sentiment was shared across the audience rather than driven by a small group of reviewers.

Key Observation

Although ratings covered the entire 1–10 scale, negative opinions remained a minority. Furthermore, viewers with moderate ratings wrote the longest reviews, suggesting that mixed opinions required greater justification than strongly positive or negative ones.

Final Statement

"3 Idiots" achieved broad audience approval, strong consensus, and sustained viewer engagement.

Thank You!

"The data has told its story, now I'm ready for yours."

The floor is open for Questions & Answers

*"Good analysts calculate data.
Great analysts explain data beautifully. "*

Presented by: Kingshuk Goswami